ADS Experiment 1

AIM: To explore **descriptive and inferential statistics** of the Customer Segmentation dataset and visualize the data using statistical plots.

THEORY:

1. Descriptive Statistics

Descriptive statistics summarize and describe the main features of a dataset.

1. Mean

The mean is the average value.

Formula:

$$\boldsymbol{Mean = \frac{\sum x_i}{n}}$$

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2. Median

The median is the middle value after sorting the data.

Example

Data = 10, 15, 20, 30, 40
Median = 20

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3. Mode

The mode is the value that appears most frequently.

Example

Data = 2, 3, 3, 5, 6
Mode = 3

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4. Range

Difference between maximum and minimum value.

$$\boldsymbol{Range = Max - Min}$$

Example

Max = 80
Min = 20

Range = 60

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5. Quartiles

Quartiles divide data into four equal parts.

- **Q1 = 25th percentile**
- **Q2 = Median**
- **Q3 = 75th percentile**

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6. Interquartile Range (IQR)

$$IQR = Q3 - Q1$$

Measures spread of middle 50% of data.

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7. Variance

Variance measures how far values spread from the mean.

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{n}$$

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8. Standard Deviation

Square root of variance.

$$\sigma = \sqrt{\sigma^2}$$

It measures dispersion.

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9. Correlation

Correlation measures relationship between two variables.

Range:

- **+1 → Perfect positive correlation**
- **0 → No correlation**
- **−1 → Perfect negative correlation**

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10. Standard Error of Mean (SE)

$$SE = \frac{\sigma}{\sqrt{n}}$$

Measures accuracy of sample mean.

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11. Coefficient of Variation

$$CV = \frac{StandardDeviation}{Mean}$$

Measures relative variability.

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12. Skewness

Measures asymmetry of distribution.

- **Positive skew → long right tail**
- **Negative skew → long left tail**

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13. Kurtosis

Measures "peakedness" of distribution.

- **High kurtosis → sharp peak**
- **Low kurtosis → flat distribution**

Inferential Statistics

Inferential statistics allow conclusions about populations using samples.

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1. Distributions

Normal Distribution

A symmetric bell-shaped distribution.

Properties:

- **Mean = Median = Mode**
- **Most values near mean**

Example: Heights of people.

Poisson Distribution

Used for counting events occurring in fixed intervals.

Example:

Number of customers entering store per minute.

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2. Confidence Interval

A range of values that likely contains the population parameter.

Formula:

$$CI = \bar{x} \pm Z \times \frac{\sigma}{\sqrt{n}}$$

Example:

Mean income = 50,000
CI (95%) = 50,000 ± error

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3. Hypothesis Testing

Used to test assumptions about data.

Null Hypothesis (H0)

No difference exists.

Alternative Hypothesis (H1)

Difference exists.

Z-Test

Used when sample size > 30 and population variance known.

Formula:

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

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t-Test

Used when sample size is small and variance unknown.

Formula:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Type I Error

Rejecting a true null hypothesis.

Type II Error

Failing to reject a false null hypothesis.

ANOVA (Analysis of Variance)

Used to compare means of multiple groups.

Example:

Comparing spending score across different age groups.

Outcome and Conclusion:

1. Customer Segmentation Foundation

The descriptive statistics reveal important spending patterns through variables such as UnitPrice and Quantity, along with geographic distribution through Country. These insights form the base for customer segmentation by identifying groups such as:

- **High-value customers**
- **Low-value customers**
- **Frequent buyers**
- **Occasional buyers**

Understanding these patterns helps businesses design targeted marketing strategies and personalized recommendations.

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2. Data Quality Assessment

Data quality is evaluated using the following techniques:

- **Missing Value Analysis (N Missing):**
  Helps identify incomplete records such as missing CustomerID, which must be handled before building machine learning models.
- **Outlier Detection:**
  Using Box Plots and the Interquartile Range (IQR) method helps detect abnormal transactions such as:
  - **Extremely large bulk orders**
  - **Product returns represented by negative quantities**

Detecting these anomalies is essential to ensure reliable analysis and modeling.

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3. Distribution Understanding

Statistical distribution measures provide insight into how the data behaves.

- **Skewness and Kurtosis:**
  The variables UnitPrice and Quantity are highly right-skewed, meaning most transactions are small while a few are extremely large.
- **Because of this skewness, the average (mean) may not accurately represent the data. Instead, the trimmed mean provides a more reliable central tendency.**

- **Normal Distribution Fit:**
  Visualization shows that the data does not follow a normal distribution, indicating that non-parametric statistical methods may be more appropriate for modeling.

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4. Relationship Discovery

Understanding relationships between variables helps identify behavioral patterns.

- **Correlation Matrix:**
  Shows the strength of relationships between numerical variables such as Quantity and UnitPrice. This helps in analyzing pricing strategies and demand patterns.
- **Scatter Plot:**
  Visually examines whether higher purchase quantities correspond to higher or lower unit prices.

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5. Statistical Validation

Several inferential statistical methods validate the findings.

- **Z-Test and T-Test:**
  Used to determine whether a sample of customers represents the entire population accurately.
- **ANOVA (Analysis of Variance):**
  Tests whether customers from different countries have significantly different spending patterns (UnitPrice). This helps determine if geographic segmentation is meaningful.
- **Confidence Intervals:**
  Provide a range within which the true population parameter is likely to lie, helping quantify uncertainty in sample estimates.

Key Findings and Conclusions

Finding	Conclusion
Skewness > 0 for Quantity and UnitPrice	Data is right-skewed; most transactions are low-value with a small number of high-value purchases
High Kurtosis	The dataset contains heavy tails, meaning extreme values occur more frequently than in a normal distribution
High Coefficient of Variation (CV%)	Large variability in customer spending indicates diverse purchasing behavior, making segmentation important

Finding	Conclusion
Z-Test / T-Test Results	Sample means are not significantly different from the population mean, suggesting that random samples represent the dataset well
Type I Error Simulation	The observed error rate is close to the significance level ($\alpha = 0.05$), confirming that the statistical test is properly calibrated
ANOVA Across Countries	If the null hypothesis is rejected, spending patterns differ across countries, supporting geographic segmentation
Correlation Analysis	Weak correlation between Quantity and UnitPrice indicates they represent independent aspects of customer behavior

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Final Conclusion

The experiment demonstrates that customer purchasing behavior in the dataset is highly variable and does not follow a normal distribution. This variability indicates strong potential for customer segmentation.

Inferential statistical tests such as ANOVA confirm that geographical differences in spending patterns exist, making country-based segmentation meaningful. Meanwhile, descriptive statistics provide key metrics required to develop clustering or classification models for customer segmentation in future analysis.

Overall, the experiment establishes a strong statistical foundation for building data-driven customer segmentation and predictive analytics models.